C Lisa

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.76
DAV MODEL SCHOOL DURGAPUR	Class XII (CBSE)	2019	83.50%
DAV MODEL SCHOOL DURGAPUR	Class X (CBSE)	2017	10.00%

SKILLS

- **KiCad** - Designed multi-layer PCBs including schematic capture and layout routing for power electronics circuits.
- **LTSpice** - Simulated converter topologies and control loops to validate circuit behavior before hardware realization.
- **Altium Designer** - Developed high-density PCB layouts with impedance-controlled routing for mixed-signal boards, for the PV Emulator M.Tech Thesis
- **Python** - Automated data acquisition, signal processing, and lookup-table generation for control algorithms.
- **Code Composer Studio** - Programmed and debugged TI microcontrollers for real-time control and PWM generation.
- **Overleaf** - Authored technical reports and thesis documentation using LaTeX for structured academic writing.
- **Matlab** - Used throughout M.Tech coursework and research for modeling, data analysis, and control algorithm design.
- **Simulink** - Applied throughout M.Tech for building and simulating closed-loop control across multiple courses and projects.

SCHOLASTIC ACHIEVEMENTS

Secured **First Position** in the **ASME Elevator Pitch Competition** and received a cash award of USD 250.

Awarded **Best Orator** twice during school for excellence in public speaking and communication.

POSITIONS OF RESPONSIBILITY

- **Placement Coordinator (Power Electronics & Power Systems)**, IIT Hyderabad. Coordinating with recruiters, assisting in placement activities, and facilitating communication between companies and students.
- **Organizing Committee Member of ICPS** -Anchored for technical sessions and sessions on Women in Engineering, and participant management for **International Conference on Power Systems 2025**.

RELEVANT COURSES

- Multilevel Converters
- Power Converter Design
- Embedded System Design

PROJECTS

1. **Hardware-in-the-Loop Photovoltaic Emulator with Lookup Table-Based Current Estimation and PI-Controlled Buck Converter.**

PCB Design & Hardware Development

- Designed a multi-layer PCB integrating the Buck converter, gate driver circuit, voltage/current sensing circuitry using Altium

Embedded Control & Firmware Development

- Programmed the microcontroller (Arduino) to acquire real-time voltage and current data from the sensing circuit.
- Implemented a PI control algorithm on the MCU to generate PWM signals driving the Buck converter.

PV Characteristic Emulation & System Integration

- Developed MPU-side logic to fetch Ipv values from a large PV characteristic lookup table based on sensed V/I inputs.
- Established real-time MCU-MPU communication to close the control loop for accurate PV emulation.

2.**Sun Tracking Solar Panel**

- Developed an Arduino based embedded control system for automatic dual axis solar tracking
- Programmed PWM control of servo motors for precise panel orientation.

3.**Design of high frequency inductor for boost converter .**

- Designed a high frequency boost converter inductor using the Area Product method.
- Calculated core dimensions, winding turns, air gap and conductor size.

4.**Investigation of ADC and ePWM Peripherals on TI C2000 Microcontroller.**

- Configured ADC and ePWM peripherals on the TIC2000 microcontroller for real-time acquisition and PWM generation.
- Implemented ADC-Controlled PWM duty cycle and frequency modulation using potentiometer inputs.
- Generated 50Khz PWM and verified PWM waveforms and ADC measurements using a DSO.

5.**PCB Design and Hardware Implementation of a Boost Converter**

- Designed a closed-loop boost converter with magnetic components, gate driver and sensing circuits.
- Created the schematic, PCB,BOM and Gerber files in KiCad.
- Validated hardware performance using the TIC2000 based digital control.

WORK EXPERIENCE

- **Capgemini | Salesforce Intern -** Completed 3 months internship in the Salesforce CRM, Built projects by setting up custom tables, automating tasks with flow builder, controlling user access using Profiles and Permission sets.

- **Alloy Steel Plant Durgapur -** Industrial experience in Steel Melting Shop and gained exposure to electrical systems and maintenance.